

Khushi Raj

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Profile

I am a third-year undergraduate pursuing a BS–MS degree in Physics at IISER Berhampur, with a strong focus on theoretical and quantum physics. My academic interests center on quantum technologies, particularly quantum information, quantum computation, and quantum cryptography. I am motivated to explore their theoretical foundations and applications in modern physics research.

Education

Indian Institute of Science Education and Research Berhampur

Sept 2023 – Present

BS-MS

- **Coursework and labs in Physics:** Electromagnetism, Waves and Introductory Optics, Quantum Physics, Classical Mechanics, Mathematical Methods I, Quantum Mechanics I, Thermal Physics, Optics Lab.
- **Other Coursework and labs:** Fundamental courses in Chemistry, Mathematics, Biology, Earth and Environmental Science, Computer and Data Science, Humanities and Social Science
- CPI: 8.0/10 – Latest SPI: 9.3/10

Other Courses and Certifications

Machine Learning Specialization

Prof. Andrew Ng

- Supervised Learning: Regression, classification, and evaluation metrics (precision, recall, F1 score)
- Advanced Algorithms: Neural networks, decision trees, ensemble methods, and regularization
- Unsupervised Learning: Clustering (K-means, GMM), PCA, and recommendation systems
- Reinforcement Learning: Basics of MDPs and Q-learning.

Internship

Prof. Utpal Roy (IIT Patna, Summer 2025)

- During my summer internship, I studied the foundational concepts of quantum computation by thoroughly going through the first four chapters of Nielsen and Chuang's textbook. I then gained practical experience by learning Qiskit and implementing various quantum protocols. Additionally, I explored quantum machine learning, focusing on amplitude encoding techniques. The major part of my internship involved an in-depth study of research papers on Quantum Key Distribution (QKD), culminating in several presentations where I communicated key findings and insights.

Seminars Attended

- **Mathematics for Data Security, IIT Madras (May 2025):**
 - Introduction to Lattice Based Cryptography
 - Decentralization, Distributed Trust, and Blockchain
 - Distributed Graph Algorithms and Congestion
 - Factoring, RSA Security, and Lattice Attacks
- **Physics Symposium at IISER-BPR (October 2023)**

Skills

Programming Languages: Python, MATLAB, JavaScript, Qiskit

Tools and Technologies: Git and Github, Jupyter, SQL, MongoDB, PHP

Languages: English, Hindi